

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING COURSE
CODE: CSA4305

COURSE OUTCOME COVERED

CO 4: Formulate data-driven web solutions using XML, XSLT, and JSP within the MVC framework.
(L4)

Assessment Tool: Class Test

Weightage: 10%

QUESTIONS:

Total Marks: 20

- Explain the fundamental concepts used in web development by defining the following terms with suitable explanations: **XML, JSP, MVC architecture, XSLT, XML Schema, JSTL, PHP, Database Connectivity, and Parser**. Your answer should clearly describe the purpose and role of each concept in web applications.
- Explain how the above technologies are interconnected in building a web application. Describe how XML is structured and validated, how JSP and PHP are used for server-side processing, how MVC architecture organizes the application, and how database connectivity and parsers contribute to data processing and communication.

Code of Conduct:

I Aaron Samuel S(192411022)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understanding	Basic understanding with some errors	Incorrect or very limited understanding
Content Coverage	25%	Covers all required topics comprehensively	Covers most topics with minor omissions	Covers some topics only	Very few topics covered
Technical Accuracy	20%	All explanations are technically correct and precise	Minor technical errors	Some incorrect or vague explanations	Major technical errors
Integration of Concepts	15%	Effectively connects and relates concepts logically	Some connections made with minor gaps	Limited connections between concepts	No clear relationship between concepts

1 Fundamental Concepts in Web development.

* XML →

This is used to store, organize and exchange data in a structured format.

It uses custom tags to describe information

* JSP →

This is a Java based server-side tech. used to create dynamic web pages.

It allows Java code to work in HTML.

* MVC architecture:-

Divides the application into Model, View, & Controller. The model handles data the view display info. and the controller handles user requests

* XSLT:-

Used to transform XML data into format such as HTML, XML or text, making XML data easier to display.

* XML Schema:-

Defines the structure & rules of a doc. & checks whether the data is valid.

* PHP:-

It's a server side scripting lang. used to create dynamic websites and handle forms, sessions & databases.

* Database connectivity:-

It allows a web application to connect with a database to store, retrieve, update and delete data.

* Parser:-

A parser reads analysis structured data such as XML & makes info available for processing by the application.

2. How these technologies are interconnected.

→ These tech. work together to build a complete web app. XML stores and transfer data while XML Schema validates its structure. A parser reads the XML data and XSLT can transform it into HTML for display.

→ JSP or PHP perform server-side processing and generates dynamic web pages. In JSP applications, JSTL simplifies common tasks such as loop and conditions.

→ The MVC architecture organizes the application into Model View and controller. The model manages data and database operations, the view presents info to users & the controller handles requests.

→ Database connectivity connects the application to databases, allowing data to be stored & retrieved. Together, these tech. provide a complete process for data storage, validation, processing, communication & presentation in web app.